

Spring 2025: CSCI 181RT

Real-Time Systems in the Real World

Lab 2

Wednesday, February 12, 2025

Edmunds Hall 105

1:15 PM - 4:00 PM

Professor Jennifer DesCombes

Agenda

- Project Structure
- Goals

Project Structure

- Application Builder / Tool Generated Files
 - Shell
 - OS / RTOS
 - Try to Avoid Editing These Files
- User / Project Generated Files
 - Type Files
 - Global Declaration Files
 - Device / IO Support Files
 - Processing Files
 - Everything Else Needed to Do the Work
 - Tasks

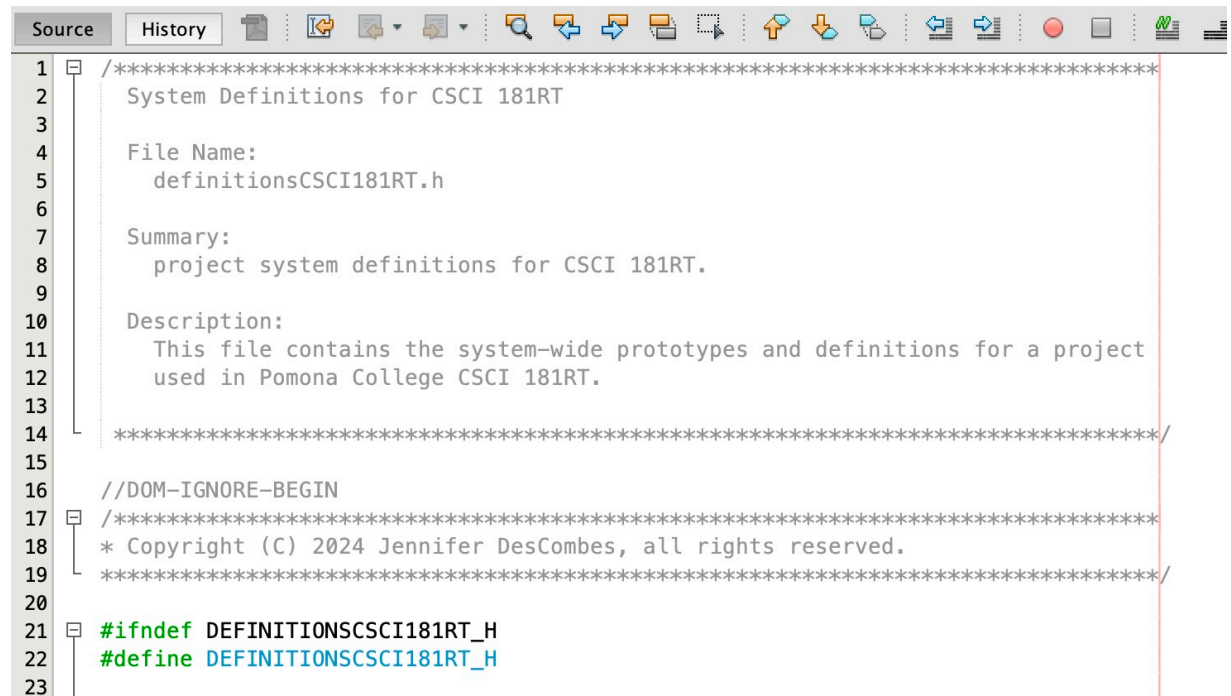
Project Structure

- Application Builder
/ Tool Generated
Files

```
Source History
1  /*****
2  System Definitions
3
4  File Name:
5  definitions.h
6
7  Summary:
8  project system definitions.
9
10 Description:
11 This file contains the system-wide prototypes and definitions for a project.
12
13 *****/
14
15 //DOM-IGNORE-BEGIN
16 /*****
17 * Copyright (C) 2018 Microchip Technology Inc. and its subsidiaries.
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36 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
37 *****/
38 //DOM-IGNORE-END
```

Project Structure

- User / Project Generated Files



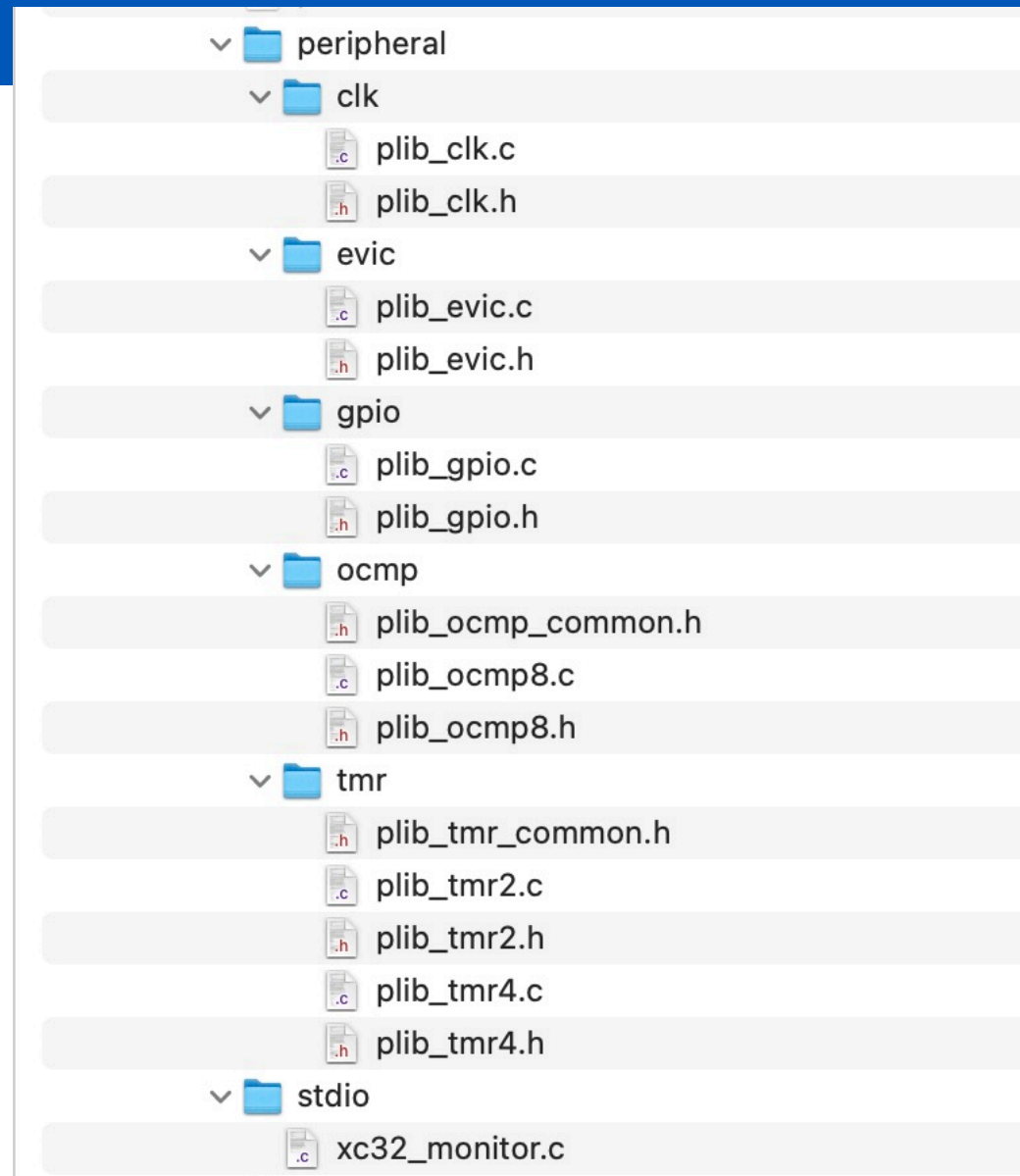
```
1  /*****
2  System Definitions for CSCI 181RT
3
4  File Name:
5  definitionsCSCI181RT.h
6
7  Summary:
8  project system definitions for CSCI 181RT.
9
10 Description:
11 This file contains the system-wide prototypes and definitions for a project
12 used in Pomona College CSCI 181RT.
13
14 *****/
15
16 //DOM-IGNORE-BEGIN
17 /*****
18 * Copyright (C) 2024 Jennifer DesCombes, all rights reserved.
19 *****/
20
21 #ifndef DEFINITIONSCSCI181RT_H
22 #define DEFINITIONSCSCI181RT_H
23
```

Project Structure

- Physical Grouping of Files
 - One Folder for Each Defined Object
 - Types
 - IO / Device Support
 - Tasks
 - Each Folder Should Contain
 - Code (.c)
 - Header (.h)
 - Documentation
 - Test Files
 - Other Support Files or Data

Project Structure

- Physical Grouping of Files
 - Headers and Code in Same Folder

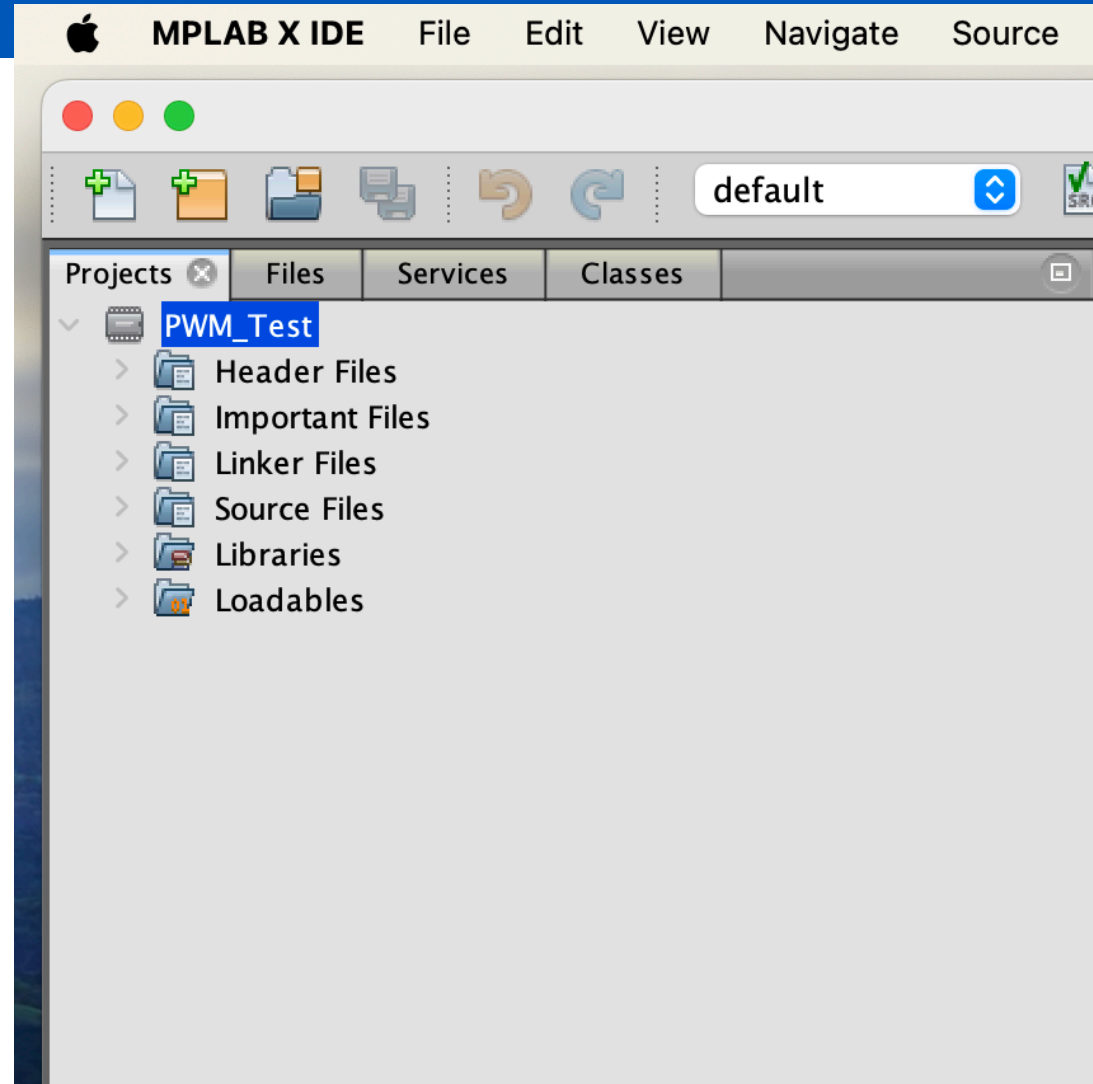


Project Structure

- Logical Grouping of Files Within IDE (Integrated Development Environment)
 - Will Vary from IDE to IDE
 - Typically Will List All Headers Together
 - Typically Will List All Code Together
 - Typically Will Allow Construction of Virtual Folders Within These Groups to Provide Functional Grouping of Code

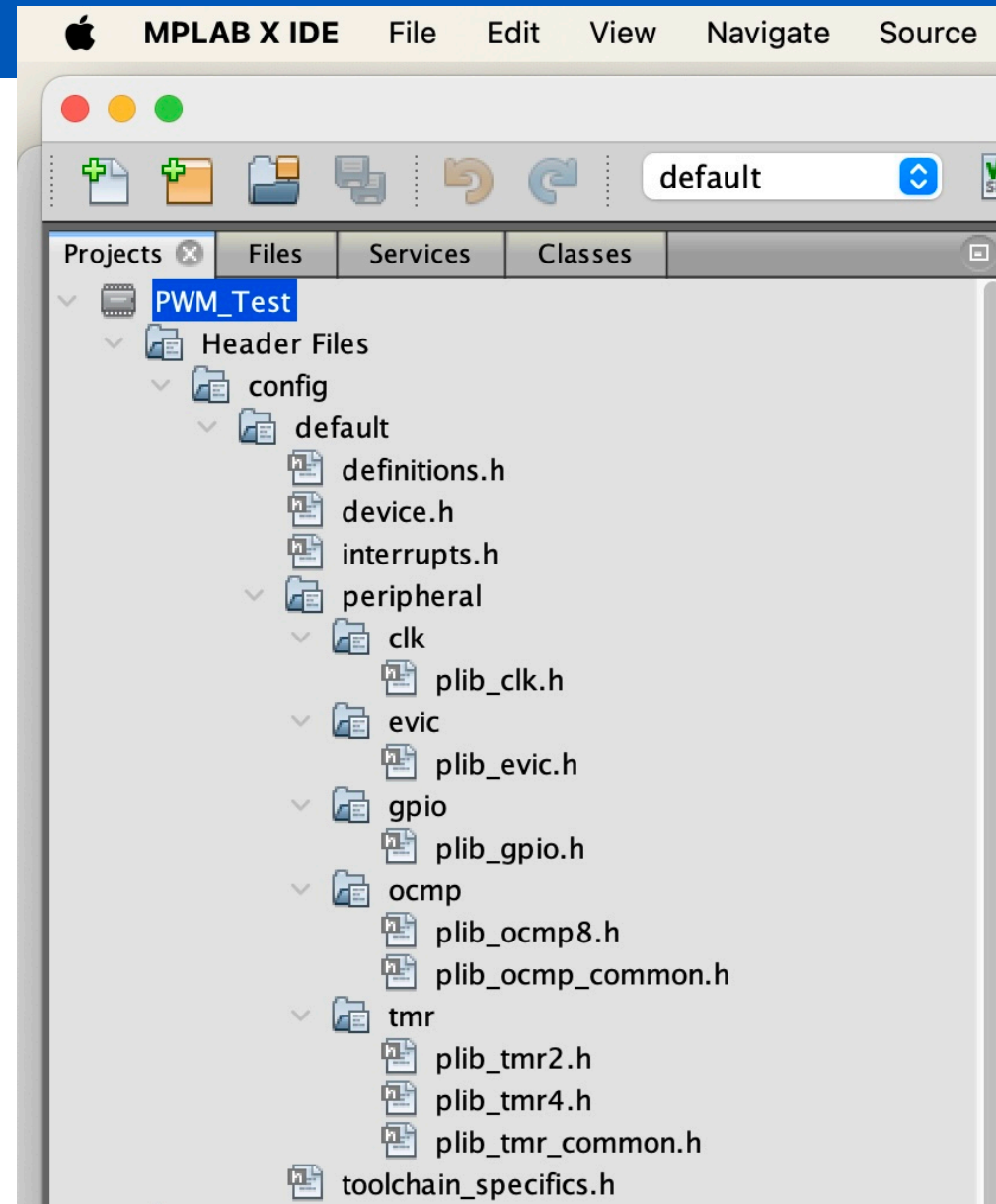
Project Structure

- IDE Grouping of Files



Project Structure

- IDE Grouping of Files
 - Headers



Project Structure

- IDE Grouping of Files
 - Code



Lab 2 Goals - Primary

- Complete Download of Tools
 - Can Use Any Resources Available
 - Can Use Each Other as Resource (Collaboration)
- Establish Communications from Evaluation Board to Computer
- Compile and Load Sample Program (provided)
- Using Sample Program as a Framework, Write, Compile and Execute “Hello World” Program
 - Display in Terminal Window

Lab 2 Goals - Secondary

- Add Character Input from Terminal
 - 'H' or 'h' - Redisplay "Hello World"
 - Include CR and LF?
 - Character Input Without Carriage Return (Enter Key)
- Use Character Input from Terminal to Turn on LEDs
- Use Character Output to Terminal Indicate State of Buttons

Assignment - Lab Preparation

Documentation

Title		
Curiosity PIC32MZ EF 2.0 Development Board Users Guide	Download	☆
PIC32MZ_EF Curiosity Board 2.0 Design Documentation	Download	☆
Create a new MPLAB Harmony v3 project using MCC	Link	
Update and Configure an Existing MHC-based MPLAB Harmony v3 Project to MCC-based Project	Link	
Getting Started with Harmony v3 Peripheral Libraries on PIC32MZ EF MCUs	Link	
Getting Started with Harmony v3 Drivers and Middleware on PIC32MZ EF MCUs using FreeRTOS	Link	
Creating a Hello World Application on PIC32 Microcontrollers Using MPLAB Harmony v3 and the MPLAB Code Configurator (MCC)	Download	☆
How to Build an Application by Adding a New PLIB, Driver, or Middleware to an Existing MPLAB Harmony v3 Project	Download	☆
How to Use the DMA CRC Generator on PIC32MX/PIC32MZ/PIC32MM Devices	Download	☆

NOTE: Date reflects original presentation date.

Lab 2 Goals - Secondary

- PIC32MZ Embedded Connectivity with FP Unit (EF) Family

<https://ww1.microchip.com/downloads/aemDocuments/documents/MCU32/ProductDocuments/DataSheets/PIC32MZ-Embedded-Connectivity-with-Floating-Point-Unit-Family-Data-Sheet-DS60001320H.pdf>